

EDUCATION

Tulane University

B.S. in Mathematics, GPA: 3.97 / 4.00

New Orleans, LA

August 2024 – May 2027, Expected

RESEARCH INTERESTS

- Post-quantum and code-based cryptography; algebraic coding theory; algebraic geometry in cryptography.

PUBLICATIONS & PREPRINTS

- [1] M. B. Can and E. C. Naig, “A Theory of Leakage in Secret Sharing and Evaluation Codes”, Submitted to *Advances in Mathematics of Communications*, under review, 2026.
- [2] M. B. Can and E. C. Naig, *Applications of Bruhat-Chevalley-Renner Decomposition to Metric-Aware Code-Based Cryptography*, Cryptology ePrint Archive, Paper 2026/665, Preprint, 2026. [Online]. Available: <https://eprint.iacr.org/2026/665>
- [3] M. B. Can and E. C. Naig, “Module-HQC”, Submitted to IEEE Transactions on Information Theory, 2026.
- [4] M. B. Can, S. Chakrabartty, and E. C. Naig, “Post-Quantum Key Establishment on a Self-Powered Timekeeping Device”, Submitted for journal publication, 2025.
- [5] M. B. Can and E. C. Naig, “Quasi-Circulant Trapdoor Construction for Niederreiter-Type Key Encapsulation”, Submitted to *Designs, Codes and Cryptography*, under review, 2025.

PATENTS

- **Systems and Methods for Secure and Reliable Data Storage and Transmission** Filed April 2026
PCT patent application; co-inventors: Mahir Bilen Can, Dillon Montero, and Eli Coe Naig. Tulane Ref. No. 2025-010-02; Foley Ref. No. 136669-0184.
- **Metric-Aware Code-Based Cryptography** Filed March 6, 2026
U.S. provisional patent application; co-inventors: Mahir Bilen Can and Eli Coe Naig. Attorney docket no. 752001-8010.

RESEARCH EXPERIENCE

Undergraduate Researcher

Advisor: Mahir Bilen Can, Tulane University

New Orleans, LA

August 2025 – Present

- Investigate algebraic and metric-based code families for post-quantum cryptography, emphasizing decoding hardness, structural security, and non-Hamming metrics.
- Develop Niederreiter-type quasi-cyclic and quasi-circulant constructions and analyze performance and security under NRT metrics.
- Derive bounds and resistance criteria using linear algebra, group actions, parity-check methods, and algebraic geometry.

- Coauthor manuscripts in code-based cryptography, secret sharing, and metric-aware coding theory; participated in a Tulane Innovation Institute initiative on research translation and proposal development.

NSF REU in Quantum Information Science

University of South Carolina

Columbia, SC

May 2026 – July 2026

- Selected participant in the U.S. National Science Foundation REU program award 2447817.
- Selected to conduct research in quantum information science with Stephen Fenner, Peng (Frank) Fu, and Homayoun Valafar.

RTG Summer Math Program

Louisiana State University

Baton Rouge, LA

May 2025 – June 2025

- Participated in the LSU RTG Summer Math Program in Combinatorics and Geometry, focusing on hyperplane arrangements and matroid theory.
- Developed and delivered an advanced lecture series on arrangements, geometric lattices, Möbius functions, and characteristic polynomials.
- Led peer discussions connecting combinatorial geometry to topology, coding theory, and cryptography.

TECHNICAL & ENTREPRENEURIAL EXPERIENCE

QSPARC Labs

Co-Founder and CEO

New Orleans, LA

August 2025 – Present

- Co-founded a post-quantum security venture translating algebraic coding theory into secure communication and data-protection systems.
- Architect reference implementations and message-protection workflows in Python, Rust, and C.
- Build validation and performance-testing harnesses for cryptographic primitives under realistic threat models.
- Lead technical roadmap, operations, research-to-product execution, and intellectual property strategy, including patent development.
- Selected for an NSF-funded I-Corps-style commercialization program administered through MIT.

TEACHING & MENTORING

- Volunteer Tutor, Tulane University Fall 2025 – Present
Tutor student-athletes in linear algebra and abstract algebra, supporting coursework alongside practice and competition schedules.

AWARDS & GRANTS

- Tulane Honors Scholar 2024–2028
- Dean's List, Tulane University Fall 2024; Spring 2025; Fall 2025; Spring 2026
- CELT Faculty-Mentored Undergraduate Research Grant, Tulane University; mentor: Mahir Bilen Can. 2026
- Selected for the Newcomb-Tulane College Summer Research Program; declined to participate in NSF REU. 2026

LEADERSHIP & SERVICE

- President, Tulane Math Club 2026–2027
Lead programming and organization for the university math club.
- Co-Founder and Vice President, Tulane Science Olympiad 2024–Present
Co-founded Tulane's first Science Olympiad invitational; coordinated logistics across 60+ classrooms and laboratories for a 13-team, 450+ student competition and developed tulanescioly.org.
- Trip Leader, Tulane Outdoor Adventures 2024–Present
Lead outdoor trips for student groups, managing route planning, risk assessment, logistics, and participant safety.

TECHNICAL SKILLS

- **Programming:** Python, Rust, C, C++, JavaScript, Java, MATLAB, R, SQL
- **Tools:** Git, GitHub, L^AT_EX, Linux
- **Languages:** English (native), Chinese (intermediate reading/writing; beginner speaking)

SELECTED ADVANCED COURSEWORK & INDEPENDENT STUDY

- **Selected Advanced Coursework:** Abstract Algebra I–II, Real Analysis I–II, Linear Algebra, Probability, and Combinatorics.
- **Planned Fall 2026 Coursework:** Algebra I, Algebraic Geometry, Mathematical Foundations of Computer Security, and Honors Thesis.
- **Independent Study:** Algebraic geometry, error-correcting codes, real analysis, and manifolds; texts include Hartshorne, Huffman–Pless, Rudin, and Spivak.